

Chapter 2

MCQs

Question 1

What is the main difference between kinetic energy and potential energy?

- ☐ A) Kinetic energy is stored energy, while potential energy is energy of motion.
- ☐ B) Kinetic energy is energy of motion, while potential energy is stored energy.
- ☐ C) Kinetic energy is only found in living things, while potential energy is in objects.
- ☐ D) Kinetic energy is caused by gravity, while potential energy is caused by friction. Correct Answer: b) Kinetic energy is energy of motion, while potential energy is stored energy.

Note: No correct answer specified for this question

Question 2

What happens during a chemical reaction?

- ☐ A) Substances mix together without changing their properties.
- ☐ B) Substances transform into new substances with different properties.
- ☐ C) Substances only change their physical shape or size.
- ☐ D) Substances disappear completely into energy. Correct Answer: b) Substances transform into new substances with different properties.

Note: No correct answer specified for this question

Question 3

Which of the following best describes the role of the circulatory system in the human body?

- ☐ A) It helps you think and react by sending signals through nerves.
- ☐ B) It breaks down food into energy for the body to use.
- ☐ C) It transports oxygen and nutrients to cells via blood.
- ☐ D) It supports the body with bones and allows movement. Correct Answer: c) It transports oxygen and nutrients to cells via blood.

Note: No correct answer specified for this question

Question 4

What is the purpose of forming a hypothesis in the scientific method?

- ☐ A) To prove your experiment will work before testing it.
- ☐ B) To make an educated guess about what might happen in an experiment.
- ☐ C) To write down the final results of your experiment.
- ☐ D) To ask a random question without any research. Correct Answer: b) To make an educated guess about what might happen in an experiment.

Note: No correct answer specified for this question

Question 5

In the "Pendulum Power" experiment from the chapter, what factor most likely affects the time it takes for a pendulum to complete a swing?

- ☐ A) The color of the string used in the pendulum.
 - ☐ B) The length of the string holding the weight.
 - ☐ C) The temperature of the room where the experiment is conducted.
 - ☐ D) The type of material the weight is made of.
- Correct Answer: b) The length of the string holding the weight.

Note: No correct answer specified for this question